

10	(a) (i)	(potential =) $1.2 / (1.2 + 4.2) \times 4.5 = +1.0 \text{ V}$	A1	[1]
	(ii)	(for $V_{\text{IN}} > 1.0 \text{ V}$) $V^+ > V^-$ output (of op-amp) is +5 V or positive diode conducts giving +5 V across R or V_{out} is +5 V	B1 M1 A1	
		(for $V_{\text{IN}} < 1.0 \text{ V}$) output of op-amp -5 V / negative so diode does not conduct, giving $V_{\text{out}} = 0$ or 0 V across R	A1	[4]
	(b) (i)	square wave with maximum value +5 V and minimum value 0 vertical sides in correct positions and correct phase	M1 A1	[2]
	(ii)	re-shaping (digital) signals/regenerator (amplifier)	B1	[1]